

Pattern Printing

These problems will help you to understand nested loops very well. For all of these problems, try to think in a row and column approach. Don't attempt any problem before you have a basic understanding of nested loop. Try to map all these problems in this format. Let's think. This problem asked you to print a box, where n = 4.

So the answer will be:

For this problem, let's think like a matrix, which is n X n. And using a nested loop, think about how you can print this.

Row -> Col ↓	0	1	2	3
0	*	*	*	*
1	*	*	*	*
2	*	*	*	*
3	*	*	*	*

Now print all these problems using nested for/while loops only. Either digits or a star or letters is fine. For any n, now print this format. For this example: n = 5 (Q1)

1.

* | 1 | A
* * | 1 2 | A B
* * * | 1 2 3 | A B C
* * * * | 1 2 3 4 | A B C D
* * * * * | 1 2 3 4 5 | A B C D E

2.

* | 1 | A
* * | 1 2 | A B
* * * | 1 2 3 | A B C
* * * * | 1 2 3 4 | A B C D
* * * * * | 1 2 3 4 5 | A B C D E

3.

* | 1 | A
* * * | 1 2 3 | A B C
* * * * * | 1 2 3 4 5 | A B C D E
* * * * * * * | 1 2 3 4 5 6 7 | A B C D E F G
* * * * * * * * * | 1 2 3 4 5 6 7 8 9 | A B C D E F G H I

4.

* * * * * | 1 2 3 4 5 | A B C D E
* * * * | 1 2 3 4 | A B C D
* * * | 1 2 3 | A B C
* * | 1 2 | A B

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*           |      1           |      A

5. * * * * * |      1 2 3 4 5      |      A B C D E
   * * * *   |      1 2 3 4      |      A B C D
     * * *   |      1 2 3        |      A B C
       * *   |      1 2          |      A B
         *   |      1            |      A

6. * * * * * * * * * * |      1 2 3 4 5 6 7 8 9      |      A B C D E F G H I
   * * * * * * * *     |      1 2 3 4 5 6 7          |      A B C D E F G
     * * * * *         |      1 2 3 4 5            |      A B C D E
       * * *           |      1 2 3              |      A B C
         *             |      1                  |      A

7.  * * * * * |      1 2 3 4 5      |      A B C D E
     * * * * * |      1 2 3 4 5      |      A B C D E
       * * * * * |      1 2 3 4 5      |      A B C D E
         * * * * * |      1 2 3 4 5      |      A B C D E
           * * * * * |      1 2 3 4 5      |      A B C D E

8.      *           |      1           |      A
         * * *       |      1 2 3       |      A B C
           * * * * * |      1 2 3 4 5   |      A B C D E
             * * * * * |      1 2 3 4 5 6 7 |      A B C D E F G
               * * * * * * * * * |      1 2 3 4 5 6 7 8 9 |      A B C D E F G H I
                 * * * * * * * * |      1 2 3 4 5 6 7   |      A B C D E F G
                   * * * * *     |      1 2 3 4 5     |      A B C D E
                     * * *       |      1 2 3       |      A B C
                       *         |      1           |      A

9. * * * * * * * * * * |      1 2 3 4 5 6 7 8 9      |      A B C D E F G H I
   * * * * * * * *     |      1 2 3 4 5 6 7          |      A B C D E F G
     * * * * *         |      1 2 3 4 5            |      A B C D E
       * * *           |      1 2 3              |      A B C
         *             |      1                  |      A
           * * *       |      1 2 3           |      A B C
             * * * * * |      1 2 3 4 5       |      A B C D E
               * * * * * * * |      1 2 3 4 5 6 7   |      A B C D E F G
                 * * * * * * * * * |      1 2 3 4 5 6 7 8 9 |      A B C D E F G H I

10.      *           |      1           |      A
          * *       |      1 3       |      A C
            * * *   |      1 5       |      A E
              * * * * |      1 7       |      A G
                * * * * * * * * * |      1 2 3 4 5 6 7 8 9 |      A B C D E F G H I
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